

Modern Era Developments and Discoveries

Photography (1827)

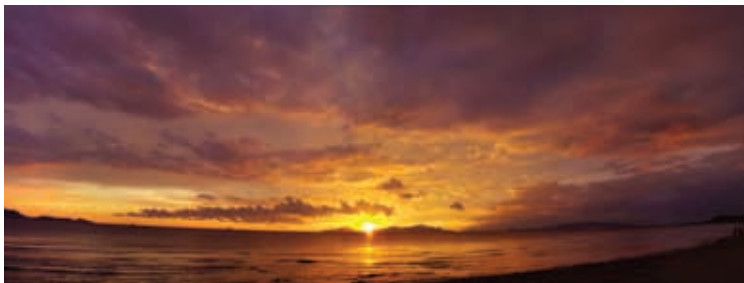
One of the earliest examples of Nanotechnology from the Nineteenth century is photography, which depends on the production of silver nanoparticles sensitive to light. Photographic film is a thin layer of gelatin containing silver halides and a base of transparent cellulose acetate. The light decomposes the silver halides, producing nanoparticles of silver, which are the pixels of the photographic image. Although the British scientists Thomas Wedgewood and Sir Humphry Davy were the pioneers in this field, the first successful photograph was produced in 1827, by Joseph Niepce using material that hardened on exposure to light. His partner Louis Daguerre discovered a way of developing photographic plates in 1839 and also to make permanent images by immersing it in salt.

Michael Faraday (1857)

Considered by many to be one of the greatest experimentalists who ever lived, the English chemist and physicist, Michael Faraday discovered and prepared the first metallic colloids (fine particles that suspend in a solution) in 1856. Faraday's gold colloids had special electronic and optical properties, and are now known as one of the many interesting metallic nanoparticles. His "ruby" gold colloid, demonstrated that nanostructured gold under certain lighting conditions produces different-coloured solutions.

MIE Theory(1908)

The MIE theory of light scattering, as proposed by the German physicist, Gustav Mie explains how different sized particles create the colours we see in the sky. His theory shows that light scatters from particles more efficiently at short wavelengths than at long wavelengths. This is the reason we perceive the sky as blue during the day and as a mixture of red and yellow during sunset. He unwittingly



Light Scattering Phenomenon